SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0704

COURSE OUTCOME COVERED

CO5:
Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

ANSWERS

Q1. Rural Branch Office – Internet Connectivity and Security

Problem Understanding and Constraints

The multinational company is establishing a branch office in a **remote rural location** where high-speed wired Internet is unavailable. The network must support:

- Cloud-based applications
- Video conferencing
- Secure communication with headquarters
- Uninterrupted business operations
- Limited connectivity budget
- Future expansion and scalability
- Reliable and secure Internet access

Therefore, the solution must balance **bandwidth, reliability, security, cost, and scalability**.

Problem Decomposition

The problem can be divided into four sub-problems:

1. **Internet Connectivity** – Select a suitable connection for the rural area.
2. **Network Infrastructure** – Select appropriate routers, switches and wireless devices.
3. **Security** – Protect communication and company resources.
4. **Reliability and Scalability** – Maintain connectivity during failures and support future users.

Recommended Solution

I recommend a **hybrid satellite/4G/5G connectivity solution**, depending on availability.
Primary connection: Satellite broadband
Backup connection: 4G/5G fixed wireless connection
A dual-WAN router can automatically switch from the primary connection to the backup connection if the main link fails.

Network Design

Internet → Dual-WAN Router → Firewall → Managed Switch →

- Employee computers
- Wi-Fi Access Points
- IP Phones
- Printers
- Local servers

The router establishes a **site-to-site VPN** with the company's headquarters.

Networking Devices

Device	Purpose
Dual-WAN Router	**Connects primary and backup Internet links**
Next-Generation Firewall	**Controls and filters network traffic**
Managed Switch	**Connects office devices efficiently**
Wi-Fi 6 Access Point	**Provides wireless connectivity**
VPN Gateway	**Provides encrypted HQ communication**
UPS	**Keeps networking equipment running during short power failures**

Security Mechanisms

1. **Site-to-site IPsec VPN** – encrypts communication between branch and headquarters.
2. **Firewall** – blocks unauthorized incoming and outgoing traffic.
3. **WPA3 Enterprise** – secures wireless communication.
4. **VLAN segmentation** – separates employees, guests and IoT devices.
5. **Multi-Factor Authentication (MFA)** – protects cloud applications.
6. **IDS/IPS** – detects and blocks suspicious network activity.
7. **Access control policies** – employees receive only required permissions.

Constraint-Based Justification

Bandwidth:

Satellite or 4G/5G can provide sufficient bandwidth for cloud applications and normal video conferencing. Traffic prioritization using **QoS** can give video conferencing higher priority.

Reliability:

Using two different Internet technologies provides redundancy. If the primary connection fails, the router switches to the backup link.

Cost:

Instead of expensive dedicated leased lines, the company can use commercially available satellite/fixed-wireless connectivity with a lower initial infrastructure cost.

Security:

VPN, firewall, VLANs, MFA and IDS/IPS protect company data despite using wireless/satellite Internet.

Future Scalability:

A managed switch, VLAN-capable firewall and dual-WAN router allow additional employees, access points and devices to be added later.

Innovative Feature

A **cloud-managed network with automatic WAN failover and QoS** can continuously monitor connection quality and automatically select the better available link.

Final Recommendation

The most feasible solution is a **primary satellite/fixed-wireless connection with 4G/5G backup, dual-WAN routing, firewall, VPN, VLAN segmentation, WPA3 and QoS**. This provides a practical balance between **limited budget, reliability, security, bandwidth requirements and future scalability**.

Conclusion:
This design allows the rural branch to operate reliably without requiring expensive wired infrastructure while maintaining secure communication with headquarters.

Q2. University Campus Wi-Fi for 3,000+ Students

Problem Understanding and Constraints

The university needs campus-wide wireless connectivity for **more than 3,000 students** in classrooms, laboratories, libraries and hostels. The major constraints are:

- Very high number of simultaneous users
- Limited number of Access Points (APs)
- Large geographical coverage
- Signal interference
- High bandwidth requirement
- Secure authentication

- Budget limitation
- Reliable performance

The solution must maximize **coverage and capacity** without simply increasing the number of APs.

Problem Decomposition

The problem can be divided into:

1. **Coverage planning** – determine appropriate AP locations.
2. **Capacity planning** – support thousands of simultaneous users.
3. **Frequency planning** – minimize interference.
4. **Authentication** – provide secure user access.
5. **Performance management** – control bandwidth and roaming.
6. **Cost optimization** – achieve maximum performance with limited APs.

Recommended Wireless Architecture

I recommend a **centrally managed Wi-Fi 6/6E network** using controller-based Access Points.
Basic architecture:

Internet → Core Router → Firewall → Core Switch → Wireless Controller → Wi-Fi 6 APs → Students/Staff

Access Point Placement

APs should **not simply be placed at equal distances**. Placement should be based on user density and building structure.

Location	AP Strategy
Classrooms	High-density placement
Laboratories	Higher capacity APs
Library	High-density APs with controlled power
Hostels	Distributed APs by floors/blocks
Corridors	Avoid excessive AP deployment
Outdoor areas	Outdoor-rated APs where required

High-density locations such as lecture halls, laboratories and libraries should receive priority because many users connect simultaneously.

Frequency Band Selection

5 GHz:
Primary band for high-speed student devices because it provides more usable non-overlapping channels and generally less interference than 2.4 GHz.
2.4 GHz:
Used mainly for older/IoT devices and areas requiring better range.
6 GHz:
If Wi-Fi 6E equipment and compatible client devices are affordable and supported, it can be used for high-capacity areas.

Interference Reduction

To minimize interference:

- Perform a wireless site survey.
- Use proper channel planning.
- Avoid unnecessary overlapping channels.
- Reduce AP transmit power where appropriate.

- Use band steering to move capable devices to 5/6 GHz.
- Enable automatic channel management.
- Avoid placing APs too close together.

Authentication and Security

The university should use:
WPA3-Enterprise + 802.1X + RADIUS authentication
Students can authenticate using university credentials.
Separate networks/VLANs can be created for:

- Students
- Faculty
- Administration
- Guests
- IoT devices

A guest network should have Internet-only access and should not be able to access internal university resources.

Performance Optimization

Because the number of APs is limited, the objective should be **capacity optimization rather than maximum signal coverage**.
The network can use:

- Wi-Fi 6 OFDMA
- MU-MIMO
- Band steering
- Airtime fairness
- QoS
- Load balancing
- Fast roaming

OFDMA allows an AP to efficiently serve multiple devices simultaneously, making it particularly useful in high-density environments.

Constraint-Based Justification

Coverage:
Strategic AP placement provides coverage in classrooms, labs, libraries and hostels.

Performance:
Wi-Fi 6, 5/6 GHz, OFDMA, MU-MIMO and load balancing improve performance for large numbers of simultaneous users.

Interference:
Channel planning, transmit-power optimization and band steering reduce interference.

Security:
WPA3-Enterprise and 802.1X provide individual authentication instead of using one common password.

Cost:
Instead of deploying a very large number of inexpensive APs, limited high-capacity APs are strategically deployed in high-density areas.

Innovative Feature

A **density-aware wireless controller** can dynamically adjust AP transmit power, channels and client distribution based on current user load.
For example:
Low load → normal AP operation
High load → load balancing + channel optimization + QoS
This makes better use of the limited AP budget.

Final Recommendation

A centrally managed **Wi-Fi 6/6E network with strategically positioned high-capacity APs, 5 GHz as**

the primary band, 2.4 GHz for legacy devices, optional 6 GHz, WPA3-Enterprise/802.1X authentication, VLAN segmentation and dynamic load balancing is the most efficient solution.

Conclusion:
The design balances **coverage, capacity, security, interference control and cost** while remaining scalable for future increases in student population.

Q3. Financial Institution – Secure Internal Network

Problem Understanding and Constraints

The financial institution handles **sensitive customer information and online banking services**, making cybersecurity extremely important.
The major constraints are:

- High security requirement
- Protection of customer information
- Secure employee access
- Online banking connectivity
- Compliance with security policies
- Minimal performance degradation
- Limited increase in operational cost
- Detection of suspicious activities
- Future scalability

Therefore, the architecture should follow a **defense-in-depth** approach.

Problem Decomposition

The problem can be divided into:

1. **Network segmentation** – isolate critical resources.
2. **Firewall protection** – control traffic between zones.
3. **Authentication** – verify users and devices.
4. **Intrusion detection** – detect attacks.
5. **Secure Internet access** – protect online services.
6. **Monitoring and compliance** – maintain logs and audit trails.

Proposed Secure Network Architecture

A suitable architecture is:
Internet
↓
Edge Router
↓
Perimeter Firewall
↓
DMZ
↓
Internal Firewall
↓
Core Switch
↓
VLANs

- Employee VLAN
- Customer-data/application VLAN
- Server VLAN
- Management VLAN
- Guest VLAN

Network Segmentation

The network should use **VLANs and firewall-based segmentation**.

For example:

Network Segment	Purpose
Employee VLAN	Normal employee systems
Server VLAN	Internal applications
Customer Data VLAN	Highly sensitive customer information

The **customer-data network must not be directly accessible from the guest or normal employee network**.

Firewall Deployment
Use a **Next-Generation Firewall (NGFW)** at the Internet boundary.
It should provide:

- Stateful packet inspection
- Application filtering
- Intrusion prevention
- URL filtering
- VPN support
- Access-control policies
- Logging

For critical financial systems, an additional internal firewall can separate the **server/customer-data zone** from normal employee systems.
Authentication

Use:
Multi-Factor Authentication (MFA) for sensitive resources.
Authentication can combine:

1. Username/password
2. One-time password/security token
3. Device authentication or certificate

Role-Based Access Control (RBAC) should ensure that employees can access only the resources required for their jobs.
For example:
Teller → Customer transaction systems
IT Administrator → Network management
HR → Employee records
This follows the **principle of least privilege**.

Intrusion Detection and Prevention

Deploy **IDS/IPS** to monitor network traffic.
IDS: Detects suspicious traffic and generates alerts.
IPS: Detects and automatically blocks known malicious traffic.
The system should monitor:

- Port scanning
- Brute-force attempts
- Malware traffic
- Unauthorized access
- Abnormal data transfers
- Suspicious login activity

Secure Online Banking Access

Public-facing banking services should be placed in a **DMZ** rather than directly inside the internal network.
Example:
Internet → Firewall → DMZ Web Server → Application Layer → Secure Database Network
The database should not be directly reachable from the Internet.

Monitoring and Logging

A centralized logging/SIEM system can collect:

- Firewall logs
- Authentication logs
- IDS/IPS alerts
- Server logs
- Access records

Security administrators can analyze these logs to detect unusual activity.

Constraint-Based Justification

Security
Multiple layers of protection are used:
Firewall + VLAN + DMZ + MFA + RBAC + IDS/IPS + encryption + monitoring
Therefore, compromising one system does not automatically expose the entire organization.

Performance
Security mechanisms should be deployed strategically rather than inspecting unnecessary traffic repeatedly.
For example:

- Use VLANs to reduce broadcast traffic.
- Apply firewall rules only where required.
- Use hardware-accelerated firewall infrastructure where appropriate.
- Prioritize critical financial applications using QoS.

This provides security without unnecessarily slowing normal operations.

Compliance
Strict authentication, segmentation, access control and centralized logging help the organization demonstrate that sensitive information is being appropriately protected and that access can be audited.

Budget
Instead of purchasing many separate security appliances, an **NGFW with integrated security features** can provide firewall, VPN, IPS and application filtering capabilities.
Open-source monitoring tools can also reduce licensing costs where appropriate.

Innovation
Implement **risk-based access monitoring**.
For example:
Normal login from an approved device → Allow
Multiple failed logins → Alert
Login from an unusual location/device + sensitive data request → Require additional verification or block
This creates adaptive security without applying expensive restrictions to every employee all the time.

Final Recommended Architecture
Internet
↓
Edge Router
↓
Next-Generation Firewall
↓
DMZ — Public Banking Services
↓
Internal Firewall
↓
Core Switch
↓
VLAN Segmentation
├── Employee Network
├── Application Servers
├── Customer Data Network
├── Management Network

└── Guest Network

Security Layer:
MFA + RBAC + IDS/IPS + Encryption + SIEM/Logging

Conclusion

The proposed architecture uses **defense in depth, VLAN segmentation, DMZ, next-generation firewalls, MFA, RBAC and IDS/IPS** to protect sensitive financial information. It maintains performance through segmentation and controlled traffic policies while reducing operational cost through integrated security technologies.

This solution provides a strong balance between **security, performance, compliance, budget and scalability**, making it suitable for a modern financial institution.